

## 三. 刚体力学

转动惯量  $J$ . 回转半径  $R_G = \sqrt{\frac{J}{m}}$  角动量  $L = \vec{r} \times \vec{p} = J\omega$  合外力矩  $M = \frac{dL}{dt} = J\beta$

旋进角速度  $\Omega = \frac{M}{L \sin \theta} = \frac{mgr}{J\omega}$  ( $r$  是质心到定点的距离). 平行轴定理.  $J = J_0 + mr^2$

## 四. 狭义相对论

$$\begin{cases} x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}} \\ t' = \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} \end{cases} \Leftrightarrow \begin{cases} x = \frac{x' + ut'}{\sqrt{1 - u^2/c^2}} \\ t = \frac{t' + ux'/c^2}{\sqrt{1 - u^2/c^2}} \end{cases} \quad \begin{cases} v_x' = \frac{v_x - u}{1 - uv_x/c^2} \\ v_y' = \frac{v_y \sqrt{1 - u^2/c^2}}{1 - uv_x/c^2} \end{cases} \Leftrightarrow \begin{cases} v_x = \frac{v_x' + u}{1 + uv_x'/c^2} \\ v_y = \frac{v_y' \sqrt{1 - u^2/c^2}}{1 + uv_x'/c^2} \end{cases}$$

能动关系  $E^2 = p^2 c^2 + m^2 c^4 \Rightarrow$  总能量不变. 总动量不变.

## 五. 机械振动

$$x = A \cos(\omega t + \varphi). \quad v = \frac{dx}{dt} = -\omega A \sin(\omega t + \varphi) \quad a = \frac{d^2 x}{dt^2} = -\omega^2 A \cos(\omega t + \varphi).$$

$$x_0 = A \cos \varphi \quad v_0 = -\omega A \sin \varphi. \Rightarrow A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2}$$

简谐振动:  $\frac{d^2 x}{dt^2} + \omega^2 x = 0$  或  $\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0. \Rightarrow a = -\frac{k}{m}x, \quad b = -\frac{k}{m}\theta. \Rightarrow \omega^2 = \frac{k}{m}, \quad \omega = \sqrt{\frac{k}{m}}.$

$$\text{单摆 } \omega = \sqrt{\frac{g}{L}} \quad \text{复摆 } \omega = \sqrt{\frac{mgL}{J}}$$

求角频率/振动周期-一般化方法: ① 设伸长量, ② 求作用力 ③ 牛二方程.

(阻尼? 受阻? 真的会考吗).

振动合成  $\rightarrow$  旋转矢量法

$$\text{同频率 } A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos(\varphi_2 - \varphi_1), \quad \tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}.$$

$$\text{不同频率 } A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos((\omega_2 - \omega_1)t), \quad v_{\text{拍}} = \frac{\omega_2}{2\pi} - \frac{\omega_1}{2\pi} = \nu_2 - \nu_1$$

$$(x_2 = A_2 \cos(\omega_2 t) \quad x_1 = A_1 \cos(\omega_1 t))$$

认识李萨如图.  $\frac{\omega_y}{\omega_x} = \frac{\text{水平方向交点个数}}{\text{竖直方向交点个数}}.$  (空白处取点. 作十字. 看交点个数).

## 六. 机械波

$$y(x, t) = A \cos \omega(t - \frac{x}{v}) \quad A.$$

A.  $\xrightarrow{\quad} B$  1. A为原点, B的坐标  $x$  (有正负)  
已知. 和. 2. 判断波的传播方向.  $\begin{cases} \text{向} +x: \omega(t - \frac{x}{v}) \\ \text{向} -x: \omega(t + \frac{x}{v}) \end{cases}$

3. 即可求得 B 处振动方程.

$$\omega \cdot \frac{x}{v} = 2\pi, \quad v = \frac{\omega}{k} = \frac{\lambda}{T} \quad \text{波的频率 vs. 波源的振动频率.}$$

$$y(x, t) = A \cos \omega(t - \frac{x}{v}) = A \cos(\omega t - \frac{\omega}{v}x) = A \cos(\omega t - kx), \quad k \text{ 为角波数.}$$

$$y(x,t) = A \cos \omega(t - \frac{x}{v}) = A \cos(\omega t - \frac{2\pi}{\lambda} x) = A \cos(\omega t - kx), \quad k \text{ 为角波数.}$$

$$u = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{F}{\mu}}.$$

$$\text{能量密度 } w = \rho A^2 \omega^2 [\sin \omega(t - \frac{x}{v})]^2, \quad \bar{w} = \frac{1}{2} \rho A^2 \omega^2$$

$$\text{球面波: } I = \frac{P_0}{4\pi r^2}$$

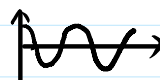
$$\text{平均能流密度(波强度)} \quad I = \bar{w} u = \frac{1}{2} \rho A^2 \omega^2 u$$

$$y(r,t) = \frac{A_0}{r} \cos \omega(t - \frac{r}{v}).$$

$$\text{波的干涉: } A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\varphi, \quad \Delta\varphi = (\varphi_2 - \varphi_1) - \frac{2\pi}{\lambda}(r_2 - r_1)$$

初相差      波程差导致的相位差

$$\text{干涉相长(相消)} \quad I \propto A^2 \Rightarrow A_{\max} = A_1 + A_2, \quad I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$



$$\text{驻波} \quad \begin{cases} y_1 = A \cos(\omega t - \frac{2\pi}{\lambda} x) \\ y_2 = A \cos(\omega t + \frac{2\pi}{\lambda} x) \end{cases} \Rightarrow y = y_1 + y_2 = 2A \cos \frac{2\pi}{\lambda} x \cos \omega t$$

$$\frac{2\pi}{\lambda} x = 2k \cdot \frac{\pi}{\lambda} x \text{ 或 } (2k+1) \cdot \frac{\pi}{\lambda} x$$

波腹      波节

- ① 原点处振动方程  $y_0, y_0$
- ② 入射波与反射波的波函数
- ③ 驻波的波函数
- ④ 波腹、波节、基频

相邻波节(波腹)间距  $\frac{\lambda}{2}$ .

固定端振幅为0  $\Rightarrow$  相位突变  $\pi$  (波程差为  $\frac{\lambda}{2}$ )  $\Rightarrow$  半波损失.

两端固定:  $y = 2A \sin \frac{2\pi}{\lambda} x \cos \omega t$  且  $y(0,t) = 0$  即  $2\pi \frac{x}{\lambda} = n\pi$

一端固定: (以固定端为原点)  $x=L$  处满足  $\frac{\partial y}{\partial x} = 0$ . 因为自由端上没有相互作用力, 绳子不作功, 即自由端是波腹.

★ 多普勒效应:  $v_R = \frac{u + v_R}{u - v_S} \cdot v_S$ , 其中  $v_R, v_S$  为相对介质质的速度. 反射(如火车进站), 称为两次计算.

## 七. 气体动理论

热力学系统: 孤立系统、封闭系统、开放系统.

$$p = \frac{1}{3} n m \bar{v}^2 = \frac{2}{3} n \bar{\epsilon} \quad \text{平均平动动能 } \bar{\epsilon} = \frac{1}{2} m \bar{v}^2 = \frac{3}{2} kT \Rightarrow \text{分子动能只与温度} T \text{ 相关, 与质量 } m \text{ 无关.}$$

$$pV = \nu RT, \quad p = nkT \quad (\text{微观, 压强 } p \text{ 只与分子数密度、温度有关}).$$

$$\text{自由度: 单原子} \rightarrow 3 \quad \text{双原子} \rightarrow 5 \quad \text{多原子} \rightarrow 6. \quad \bar{\epsilon} = \frac{i}{2} kT \quad \text{内能 } E = \nu N_A (\frac{i}{2} kT) = \nu \cdot \frac{i}{2} RT$$

$$\text{速率分布函数 } f(v) = \frac{dn}{N dv}$$

$$\text{最概然速率 } v_p = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}} \quad \text{平均速率 } \bar{v} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}} \quad \text{方均根速率 } \bar{v} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

$$\text{估算分子速率麦克斯韦速率分布函数 } f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot e^{-\frac{mv^2}{2kT}} \cdot v^2 = \frac{4}{\pi} e^{-(\frac{v}{v_p})^2} \cdot \frac{v^2}{v_p^2}$$

$$\frac{dn}{N} = f(v) dv = \frac{4}{\pi} e^{-(\frac{v}{v_p})^2} \cdot (\frac{v}{v_p})^2 \cdot \frac{dv}{v_p}$$

$$\text{等温气压公式 } p = p_0 e^{-\frac{mgh}{kT}} \Rightarrow n = \frac{p_0}{kT} e^{-\frac{mgh}{kT}} = n_0 e^{-\frac{\rho_0 gh}{kT}}$$

$$\bar{\epsilon} = \frac{1}{2} m \bar{v}^2 = \frac{3}{2} kT \quad \bar{\lambda} = \frac{\bar{v}}{n} = \frac{1}{n} \sqrt{\frac{8kT}{\pi m}} = \frac{kT}{\sqrt{2} \pi d^2 p}$$

$$\text{范德瓦耳斯方程: } [p + a(\frac{n}{V})^2] (V - nb) = \nu RT \quad \nu = 1 \text{ mol 时: } (p + \frac{a}{V^2})(V - b) = RT$$

## 八. 热力学基础.

$\Delta E = Q + A$ .  $\Delta E \rightarrow$  系统内能增量,  $Q \rightarrow$  系统吸收的热量,  $Q < 0$  放热;  $Q > 0$  吸热.

$A \rightarrow$  外界对系统做功  $(-A) \rightarrow$  系统对外做功.

对外做功  $(-A) = \int_{V_a}^{V_b} p dv$  自由膨胀 (向真空绝热膨胀)  $\Delta E = Q = A = 0$ .

定容摩尔热容  $C_{v,m} = \frac{i}{2} R$ .  $Q_v = \nu C_{v,m} \Delta T$  定压摩尔热容  $C_{p,m} = \frac{i+2}{2} R$ .  $Q_p = \nu C_{p,m} \Delta T$

Mayer 公式  $C_{p,m} = C_{v,m} + R$ . 摩尔热容比  $\gamma = \frac{C_{p,m}}{C_{v,m}} = 1 + \frac{2}{i}$

不论什么过程, 都满足  $\Delta E = \nu C_{v,m} \Delta T$ .

等温过程,  $\Delta E = Q + A = 0 \Rightarrow Q_i = (-A) = \int_{V_a}^{V_b} p dv = \nu RT \ln \frac{V_b}{V_a} = \nu RT \ln \frac{p_a}{p_b}$

绝热过程  $Q = 0$ .  $(-A) = (-\Delta E) = -\nu C_{v,m} \Delta T = \frac{\nu R \Delta T}{\gamma - 1} = \frac{\Delta(pV)}{\gamma - 1}$

泊松方程.  $pV^\gamma = \text{常量} \Rightarrow V^{\gamma-1} T = \text{常量}$ .  $p^{\gamma-1} T^{-\gamma} = \text{常量}$ .

$p$ - $V$  图中, 等温线斜率  $\frac{dp}{dV} = -\frac{p}{V}$ , 绝热线斜率  $\frac{dp}{dV} = -\gamma \frac{p}{V}$ . 交点处, 绝热线切线更陡.

热机效率  $\eta = \frac{(-A)}{Q_{\text{吸}}} = \frac{Q_{\text{吸}} - Q_{\text{放}}}{Q_{\text{吸}}}$  制冷系数  $e = \frac{Q_{\text{放}}}{A} = \frac{Q_{\text{放}}}{Q_{\text{吸}} - Q_{\text{放}}}$ .

卡诺循环.  $\eta_c = \frac{T_{\text{高}} - T_{\text{低}}}{T_{\text{高}}}$ ,  $e_c = \frac{T_{\text{高}}}{T_{\text{高}} - T_{\text{低}}}$ . (看  $Q_{\text{放}}/Q_{\text{吸}}$  是在高温热源还是低温热源).

对任意可逆过程.  $\Delta S = \int_a^b \frac{dQ}{T}$  单位:  $J \cdot K^{-1}$

绝热可逆:  $dQ = 0$ .  $\Delta S = 0$

等体可逆:  $dQ_v = \nu C_{v,m} dT$ ,  $\Delta S = \int_{T_a}^{T_b} \frac{dQ_v}{T} = \nu C_{v,m} \ln \frac{T_b}{T_a}$ .

等压可逆:  $dQ_p = \nu C_{p,m} dT$ ,  $\Delta S = \int_{T_a}^{T_b} \frac{dQ_p}{T} = \nu C_{p,m} \ln \frac{T_b}{T_a}$

等温可逆.  $dQ_i = \nu RT \ln \frac{V_b}{V_a}$ ,  $\Delta S = \nu R \ln \frac{V_b}{V_a}$

可逆相变  $\Delta S = \frac{Q}{T}$

玻尔兹曼熵.  $S = k \ln w$ .

不可逆过程可用等体、等压、等温连接.

如,  $\Delta S = \nu C_{p,m} \ln \frac{V_b}{V_a} + \nu C_{v,m} \ln \frac{p_b}{p_a}$   $\Delta$